

Capacity Boost: Data Transmission in the Unified Regime

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Abstract

Traditional theories of information transmission, both classical and quantum, assume that information is evaluated within a fixed and flat informational measure. Recent developments in the theory of nonlocal unification challenge this assumption by treating information as a conserved quantity whose observable manifestations depend on relational and nonlocal structure. In this paper, we develop a rigorous account of data transmission in what we term the *Unified Regime*, where informational observables arise as stationary quantities of a nonlocal informational action. Using Theorem A.1 of Prajnanabha, Vol. 1, Issue 1, we derive the unified (NU) entropy as a weighted logarithmic functional without assuming pointwise observer action. We show how additivity follows from the minimum interaction principle (Example 6), define unified channel capacity, and compare it with classical Shannon and quantum Holevo capacities for free-space optical (FSO) channels. The resulting framework preserves all physical channel constraints while revealing a principled capacity enhancement driven by informational geometry.

1 Introduction

Information theory has achieved remarkable success by identifying fundamental limits on reliable data transmission. Shannon’s classical theory established capacity as the maximal mutual information between channel input and output, while quantum information theory extended this result through the Holevo bound for classical communication using quantum states. Despite their differences, both theories share a foundational assumption: information is quantified using a fixed entropy functional defined on a flat informational space.

However, many physical communication channels—particularly free-space optical (FSO) channels—are dominated by multiplicative noise, long-range correlations, and observer-dependent access to information. These features suggest that the geometry of information itself may be shaped by the channel and by relational structure, rather than remaining fixed.

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The theory of nonlocal unification, developed in Prajnanabha, proposes that information is a conserved quantity whose observable flux depends on nonlocal and relational constraints. Entropy, in this framework, is not a primitive concept but an emergent scalar observable derived from a more fundamental informational action.

The objective of this paper is to reformulate data transmission theory within this unified perspective. We derive the appropriate entropy functional, define unified channel capacity, and evaluate its implications for FSO links. Throughout, we emphasize logical necessity and reduction to known theory in appropriate limits.

2 Classical and Quantum Transmission Limits

2.1 Classical Capacity

For a memoryless classical channel with input X and output Y , Shannon capacity is defined as

$$C = \sup_{p(x)} I(X; Y), \quad (1)$$

where $I(X; Y)$ is the mutual information. This result assumes that uncertainty is quantified by Shannon entropy, which is extensive and additive over independent degrees of freedom.

2.2 Quantum Capacity for Classical Information

For a quantum channel $\mathcal{N}$ used to transmit classical information, the ultimate achievable rate is given by the Holevo capacity

$$C_Q = \sup_{\{p_i, \rho_i\}} \left[S \left(\sum_i p_i \mathcal{N}(\rho_i) \right) - \sum_i p_i S(\mathcal{N}(\rho_i)) \right], \quad (2)$$

where $S(\rho)$ denotes von Neumann entropy. As in the classical case, the entropy functional is fixed a priori.

2.3 Free-Space Optical Channels

FSO channels are commonly modeled as fading channels of the form

$$Y = \sqrt{I} X + N, \quad (3)$$

where I is a random turbulence-induced intensity and N is additive noise. The statistics of I are often non-Gaussian and induce strong variability in instantaneous channel quality.

3 Nonlocal Unification and Informational Observables

3.1 Informational State and Observer Functional

In the theory of nonlocal unification, the fundamental object is the informational state $S(t)$, which encodes all informational degrees of freedom accessible to the system. Observable information is obtained through an observer functional

$$\Phi_t = f_t(S(t)), \quad (4)$$

where f_t represents the informational perspective of the observer.

Crucially, information is conserved, but its observable manifestation depends on both the evolution of $S(t)$ and the evolution of f_t .

3.2 Theorem A.1: Informational Chain Rule

Theorem A.1 (Prajnanabha, Vol. 1, Issue 1) establishes a functional chain rule for informational flux:

$$\frac{d\Phi_t}{dt} = Df_t[S(t)] \cdot \frac{dS}{dt} + \left(\frac{\partial f_t}{\partial t} \right) (S(t)). \quad (5)$$

This theorem formalizes the principle that observed informational change arises jointly from system evolution and observer evolution.

3.3 Minimum Interaction Principle

Example 6 in the same work states that at least two local consciousness instruments are required for definable informational interaction. In other words, informational dynamics are only observable when relational contrast exists.

This principle implies that independent informational degrees of freedom—those lacking relational structure—cannot generate interaction terms in observed flux.

4 Derivation of Unified Entropy

4.1 From Informational Flux to Scalar Observables

We seek scalar informational observables that are stationary under admissible informational flows. Such observables must be functionals of the informational state whose time derivative vanishes or extremizes under the chain rule of Theorem A.1.

4.2 Additivity from Independence

Let the informational state decompose as $S = S_1 \oplus S_2$, where S_1 and S_2 are informationally independent. By the minimum interaction principle, no relational structure exists between them. Therefore, the observer functional must satisfy

$$f(S_1 \oplus S_2) = f(S_1) + f(S_2). \quad (6)$$

Any cross-term would introduce observer-induced interaction, contradicting Example 6.

This additivity forces the Fréchet derivative of f to admit a diagonal kernel representation, implying that the functional admits an integral density form.

4.3 Specialization to Probabilistic States

When the informational state is represented by a probability density $p(x)$, the most general additive scalar observable consistent with Theorem A.1 is

$$\mathcal{I}[p] = \int p(x) \phi(p(x)) \lambda(x) dx, \quad (7)$$

where $\lambda(x)$ is an informational weighting induced by nonlocal structure.

The additivity and conservation constraints uniquely fix

$$\phi(p) = \log \frac{1}{p}. \quad (8)$$

4.4 Unified Entropy

The unified (NU) entropy is therefore

$$H_{\text{NU}}(X) = \int \lambda(x) p(x) \log \frac{1}{p(x)} dx. \quad (9)$$

When $\lambda(x) = 1$, the informational manifold is flat and H_{NU} reduces to Shannon entropy.

5 Unified Channel Capacity

5.1 Definition

The unified capacity of a channel is defined as

$$C_{\text{NU}} = \sup_{p(x)} I_{\text{NU}}(X; Y), \quad (10)$$

where mutual information is computed using unified entropy.

5.2 Unified Capacity for FSO Channels

For an FSO channel with instantaneous signal-to-noise ratio γI , the unified capacity becomes

$$C_{\text{NU}} = \mathbb{E}_I [\log(1 + (1 + \kappa)\gamma I)], \quad (11)$$

where κ is an effective informational curvature parameter derived from $\lambda(x)$.

6 Comparison with Classical and Quantum Capacities

At low SNR,

$$\frac{C_{\text{NU}}}{C} = 1 + \kappa, \quad (12)$$

indicating a proportional gain.

For quantum transmission of classical information over a bosonic FSO channel, the Holevo capacity takes the form

$$C_Q = \mathbb{E}_I[g(\bar{n}I)], \quad (13)$$

where $g(\cdot)$ is the bosonic entropy function. In the unified regime, the effective photon number becomes $\bar{n}(1 + \kappa)$, yielding

$$C_{\text{NU}} = \mathbb{E}_I[g(\bar{n}(1 + \kappa)I)]. \quad (14)$$

This enhancement does not violate quantum limits, as it arises from informational geometry rather than additional physical resources.

7 Discussion

The Unified Regime reframes data transmission by recognizing entropy as an emergent, observer-covariant quantity. Nonlocality enters through informational weighting and admissible flows, while additivity and locality of representation follow from independence and the minimum interaction principle.

This framework preserves all standard results in appropriate limits while extending capacity analysis to channels with strong relational structure.

8 Conclusion

We have developed a unified theory of data transmission grounded in nonlocal unification. By deriving unified entropy from Theorem A.1 and the minimum interaction principle, we established a principled basis for unified channel capacity. Application to FSO channels demonstrates a consistent capacity enhancement relative to classical and quantum baselines. These results suggest that informational geometry is a meaningful and operational extension of modern communication theory.

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